

PA 1 : TRUSTWORTHY MACHINE LEARNING

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1 INTRODUCTION

In this first practical assignment of Trustworthy Machine Learning (TML) course, we do have image classification model with some dataset too, and the goal is to determine whether to decide, whether each sample has been used in the training of the model which is also called as MIA. The evaluation part is based on the scores of TPR@5%FPR rate.

2 METHODOLOGY

2.1 OVERVIEW

The approach is simple, goal is to do a feature based membership inference attack, which means instead of directly predicting membership or class outputs from image, I first query provided target model and try to extract behavioral features from its output. The final attack combines confidence based features, uncertainty features, stability under test-time augmentation, RMIA inspired calibration, and a learned logistic regression attack model.

We use a standard image recognition model called ResNet-18, but with three small changes to fit our task. The first layer is adjusted to better handle our images, the max-pooling step is removed, and the last layer is set to predict 9 categories instead of the usual 1000. . Images are resized to 32×32 and normalized using the provided mean and standard deviation:

$$\mu = [0.7406, 0.5331, 0.7059], \quad \sigma = [0.1491, 0.1864, 0.1301].$$

As evident, Public dataset is our primary dataset that is used to train as it has the output labels present. Private dataset usage is only at the end for final evaluation and submission.

2.2 FEATURE EXTRACTION

For each sample, I run the target model and extract several features from its logits and probabilities. The first group consists of confidence based features:

I used the target model and each sample to identify some sort of behavioural pattern by using each sample and from its logits and probs, I extracted the following set of features for first group.

$$-\text{loss}, \quad p_y, \quad z_y, \quad \text{margin}, \quad \max_j p_j,$$

above, p_y is predicted p (probability) of true class, z_y is true class logit, and margin is the diff between top 2 predicted probabilities. The main advantage of extracting these features is the assumption that the model shows less loss, high confidence and large margins on training dataset sample.

My second group features consists of uncertainty based features, including negative entropy and modified entropy. It is important for detection and judgement too, because lower entropy means less uncertainty and model being more confident.

Then, finally there is also included a correctness feature, which indicates whether model predicts the true label correctly or not.

2.3 TEST TIME AUGMENTATION

To better capture prediction stability, I used test time augmentation with $T = 8$ forward passes. For each image, I evaluate the model on the original image, a horizontal flip, and slightly perturbed

versions with small Gaussian noise. The logits from these augmented views are averaged before computing the main confidence features.

In addition, I compute stability-specific features:

$$\text{mean}(p_y), \quad \text{std}(p_y), \quad \text{prediction consistency}.$$

The intuition is that other than high confidence, model should be able to make stable predictions if there has been some small changes or modifications done.

2.4 RMIA STYLE CALIBRATION

A key part of the implementation is a class-wise calibration step inspired by RMIA. Raw confidence can be misleading because some classes are naturally easier than others.

To reduce this issue, I compute the average true class probability of public non members separately for each class:

$$\bar{p}_{c,\text{nonmember}} = \frac{1}{N_c} \sum_{i: y_i=c, m_i=0} p(y_i | x_i).$$

Then, for each sample from class c , I compute score:

$$s_{\text{cal}}(x) = \frac{p(y | x)}{\bar{p}_{c,\text{nonmember}} + \epsilon}.$$

A large calibrated value means - ϵ model is more confident.

2.5 ATTACK CLASSIFIER

After feature extraction, I train a logistic regression attack model on the public dataset. The input to this model is the extracted feature vector, including the RMIA style calibrated score, and the output is the predicted probability that a sample is a member.

Standardization is done as follows, just before training:

$$x' = \frac{x - \mu}{\sigma + 10^{-8}}.$$

Classifier is trained using binary cross-entropy with logits. I use the Adam optimizer with learning rate 10^{-2} , weight decay 10^{-4} , and a maximum of 300 epochs. Early stopping with patience 40 is used to avoid overfitting.

2.6 CROSS VALIDATION AND MODEL SELECTION

I used Five fold stratified CV on public dataset. This produces out-of-fold predictions for public set and averaged predictions for private set.

Since the official metric is TPR@5%FPR, I select final scoring method using this metric directly. I compare the logistic attack output against individual features such as negative loss, true-class probability, prediction margin, negative entropy, and prediction consistency. I also evaluate linear blends:

$$s_{\text{blend}} = \alpha s_{\text{logistic}} + (1 - \alpha) s_{\text{feature}},$$

where α is searched from 0.1 to 0.9, and the score with the best public TPR@5%FPR is then applied to the private set. Finally, scores are normalized to the range $[0, 1]$ as asked in the manual and saved as `submission.csv` for final submission and evaluation.

3 RESULTS

My best public result on public dataset was:

$$\text{TPR@5\%FPR} = [0.058286].$$

The code used to reproduce the result is available at:

<https://github.com/aadrish7/Saarland-University---PA1---TML>.